

DECODED EXPLAINED

23rd July, 2026

Buried Questions: The Teesta Stage-VI Tunnel Tragedy

Buried questions

The costs of projects in environmentally fragile zones are many

India is grappling with another Himalayan tragedy. Workers in an under-construction tunnel at the Teesta Stage-VI hydroelectric project of NHPC Limited (formerly National Hydroelectric Power Corporation Private Ltd.), in Sikkim's Namchi district, struck a pocket of suspected trapped methane. The resulting burst partially damaged the tunnel, killing at least 15 of an estimated 25 workers, with rescue operations still under way. The NHPC has announced a detailed investigation. Methane is a well-recognised hazard in underground excavation, and the geology of the Teesta basin makes it a likely one. The basin sits in a seismically active zone, with young and heavily fractured rock, and where compressed pockets of gas laid down long ago can lie hidden until an excavation breaks into them. The presence of the gas, in that sense, is not in itself surprising. What remains unclear is far more consequential: the precise circumstances in which the pocket was breached, whether its risk was assessed and modelled during project planning, and whether the gas-detection and ventilation safeguards that such conditions demand were in place and functioning.

Underground work in India's difficult terrain has always been perilous. In February this year, an explosion at an illegal coal mine in Meghalaya killed about 30 workers, and in 2023, a stretch of a road tunnel under construction in Uttarakhand collapsed, trapping 41 workers for 17 days before they were pulled out. The circumstances vary, but together they underscore how lethal underground and tunnelling work remains across the country. A series of hydropower projects along the 400-kilometre stretch of the Teesta that links Sikkim, West Bengal, and Bangladesh have varied histories with environmental clearances but share a common feature – their location in the fragile Himalayan belt. That vulnerability was demonstrated tragically in October 2023, when a glacial lake outburst flood from the South Lhonak lake destroyed the Teesta-III dam and killed more than a 100 people downstream. Teesta-VI was absorbed by the public-sector NHPC Limited after its private developers, unable to afford escalating costs, went into insolvency. Relief and rescue must remain the immediate priority, and every effort to reach those still trapped is warranted. But once the emergency has passed, the government cannot treat the disaster as an isolated misfortune. It must order a thorough, independent review of whether the conditions attached to the project's environmental clearance were, in practice, strictly met.

Buried Questions: The Teesta Stage-VI Tunnel Tragedy

Teesta River

Origin:

- It originates from the Tso Lhamo Lake at an elevation of approximately 5,280 meters in North Sikkim, India
- It is glacier-fed, with sources like Pahuni Glacier, Kangse Glacier, and ChhoLhamo Lake contributing to its flow.
- It is a major tributary of the Brahmaputra River (known as the Jamuna in Bangladesh) and flows through India and Bangladesh.
- Tributaries: Major tributaries include Lachung Chhu, Dik Chhu, and Rangpo Chhu on the left bank and the Rangit River on the right bank.
- Notably, 83% of the basin lies in India, while 17% is in Bangladesh.



Buried Questions: The Teesta Stage-VI Tunnel Tragedy

With reference to river Teesta, consider the following statements:

1. The source of river Teesta is the same as that of Brahmaputra but it flows through Sikkim.
2. River Rangeet originates in Sikkim and it is a tributary of river Teesta.
3. River Teesta flows into the Bay of Bengal on the border of India and Bangladesh.

Which of the statements given above is/are correct?

This question was previously asked in

UPSC Civil Services (Prelims)
General Studies Official Paper-I...

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1. 1 and 3 only

2. 2 only

3. 2 and 3 only

4. 1, 2 and 3

India Must Lead in Securing the Strait of Bab el-Mandeb

India must lead in securing the Strait of Bab el-Mandeb

On the heels of the Strait of Hormuz crisis comes the new threat – a Strait of Bab el-Mandeb crisis. On July 16, Reuters reported that Iran had instructed the Houthis to close the Strait (which is between Yemen and the Horn of Africa) if the United States attacked Iran's power plants.

Whether the Houthis have the capacity to threaten the Strait or whether this is part of Iran's signalling to the U.S. is secondary. What is important is that this represents a risk to one-third of India's entire trade.

Revisiting 2024

We have been here before. When the Gaza war started in October 2023, the Ansarallah group in Yemen, also known as the Houthis, claimed solidarity with the Palestinians and launched drones and missiles at Israel. From November 2023 to March 2024, the Houthis targeted at least 40 vessels in the Red Sea, claiming that the vessels were linked to or bound for Israel.

In response, the U.S., together with allies including Singapore and Sri Lanka, launched Operation Prosperity Guardian (December 2023) to ensure "freedom of navigation" in the Red Sea. This was followed by United Nations Security Council Resolution 2722 (2024), which was adopted in January 2024. Israel also carried out unilateral strikes against the Houthis. The Indian Navy rescued the crew of a British tanker after it was set on fire (January 26-27, 2024). By the time the U.S declared a ceasefire in May 2025, the Houthis had been weakened, but much of the damage had already been done.

Marine insurance companies tend to be highly risk-averse. With over 90% of the world's shipping insured by just 12 protection and indemnity (P&I) clubs, shipowners have limited room to negotiate. Between November 2023 and March 2024, at least 2,000 ships avoided the Suez Canal and instead sailed around the Cape of Good



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New Delhi's leadership can help prevent another major shipping crisis in the Red Sea

Hope. This delayed critical imports to India. Global shipping routes effectively reverted to what was being done 170 years ago. Egypt lost a quarter of the Suez Canal's revenue. By July 2024, Israel's Red Sea port of Eilat had filed for bankruptcy.

This time around, the situation is more unpredictable. With U.S. forces in the region focused on Iran, it is unclear whether they will seek to secure the Red Sea as they did before.

Even in 2024, when India-bound ships were attacked by the Houthis, New Delhi had to lean directly on Tehran to bring the attacks to a halt. But that was a different time for India-Iran ties. Meanwhile, the UN Security Council is more divided today than ever before, and even if it manages to call for "freedom of navigation" in the Red Sea, it is unlikely to pass a resolution approving a force to implement that call. China is also likely to again take a unilateral approach to protecting its shipping in the Red Sea through quid pro quos with the Houthis.

If the Houthis resume attacks on shipping in the Strait, the impact on India's economy – which is already reeling from the Strait of Hormuz crisis – would be dire. While India has now ordered that no Indian mariner be deployed on ships bound to or from the Strait of Hormuz, extending the same restriction to a busy route such as the Strait of Bab el-Mandeb would not be easy and would adversely affect the safety and welfare of thousands of Indian seafarers.

What India should do

In 2024, France stayed out of a U.S.-led operation to guard its role in mediating between Hezbollah and Israel in Lebanon. Those conditions no longer exist. However, France and Japan – two countries that need shipping lanes in the Red Sea for their connectivity with Asia and Europe respectively – maintain bases in Djibouti, perched on the Red Sea. Their navies have also

conducted joint drills with the Indian Navy, which creates an opportunity for India to pre-empt a recurrence of the events of 2024.

India therefore needs to pro-actively build a naval taskforce in collaboration with France and Japan to deter attacks on shipping in the Bab el-Mandeb.

India would also do well to engage Egypt and Saudi Arabia. They had refused to join the alliance against the Houthis in 2024, given their own stance on Gaza. This time, with their economies at stake, they may not stand by. On July 20, the Houthis announced a "maritime embargo" on Saudi ships, seen as a threat against their transit through the Bab el-Mandeb. India needs to engage with these nations proactively to help secure the Red Sea, from the Suez to Bab el-Mandeb, while keeping its other partners in the region informed.

A rules-based maritime response

It needs to be emphasised that this would be a theatre-specific "Trade Protection Task Force" of navies, rather than a strategic alliance. In addition to ships and personnel, the Information Fusion Centre-Indian Ocean Region hosted by India could provide the necessary maritime domain awareness, while France and Japan could complement these efforts with on-ground logistical support. The constituent navies could agree on joint rules of engagement in response to any hostile attacks. They could provide security cover or escorts requested by commercial and civilian vessels transiting the Strait, especially those bound for Europe, India, or Japan. The task force would serve as a lasting demonstration of India's resolve to ensure "respect for maritime rules and norms" as part of Security and Growth for All in the Region (SAGAR).

India has both the responsibility and the capability to prevent the Bab el-Mandeb from becoming a literal "Gate of Tears" this time.

India Must Lead in Securing the Strait of Bab el-Mandeb

● The New Threat

- On July 16, 2026, Reuters reported that Iran had instructed the Houthis to close the Strait of Bab el-Mandeb, between Yemen and the Horn of Africa, if the United States attacked Iran's power plants
- This represents a risk to nearly one-third of India's entire trade

● Revisiting the 2024 Crisis

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- The US and allies launched Operation Prosperity Guardian in December 2023 to ensure freedom of navigation, followed by UN Security Council Resolution 2722 in January 2024; the Indian Navy rescued the crew of a British tanker set on fire in January 2024
- By the time the US declared a ceasefire in May 2025, the Houthis had been weakened, but much damage had already been done to global shipping and regional economies

India Must Lead in Securing the Strait of Bab el-Mandeb



Red Sea

The Red Sea is a major hub for international tourism, known for its diving spots and vibrant marine life.

Red Sea: It is an inlet of the Indian Ocean between Africa and Asia.

Bordering Countries: Egypt, Saudi Arabia, Yemen, Sudan, Eritrea and Djibouti.

Red Sea, connected to the Indian Ocean in the south through the Bab el Mandeb strait and the Gulf of Aden.

The northern Red Sea features the Sinai Peninsula, the Gulf of Aqaba, and the Gulf of Suez, which leads to the Suez Canal.

India Must Lead in Securing the Strait of Bab el-Mandeb

Yemen

Bordering Countries: Saudi Arabia (North), Oman (East).

Bordering Water Bodies: Red Sea (West), Gulf of Aden (South), Arabian Sea (Southeast).

Strategic location: On Bab el Mandeb, the strait linking Red Sea and Gulf of Aden, an important shipping route.

Yemen's Socotra Archipelago in Indian Ocean is a UNESCO world heritage site.

Houthis: A Zaidi Shia militant group from northwestern Yemen, the Houthis emerged in the 1990s as a rebellion against the Yemeni government. Backed by Iran, they are part of the Axis of Resistance (informal coalition of Iranian-supported militias) and target Israeli-linked ships in the Bab el-Mandeb Strait in response to the Gaza conflict.

Axis of Resistance

It is a coalition of Iranian-backed groups.

The coalition's name is said to be inspired by former US President George W Bush's use of the term 'axis of evil' referring to Iran, Iraq and North Korea.

Group members

Hezbollah (Shiite militant organization Hamas), Palestinian Sunni militant group in Gaza and Houthis (Zaydi Shia militant group, Yemen)

India Must Lead in Securing the Strait of Bab el-Mandeb



Are Pellet Guns Being Used at Protests?

Are pellet guns being used at protests?

Pellet guns are used by police and military worldwide as a non-lethal crowd-control measure. They were last known to have been used during the 2024 farmers' protest along the Punjab-Haryana border. Though police denied their use, farmer leaders had alleged that several people were injured.

EXPLAINER

Vijaita Singh

The story so far:

At least one of the 80 protesters injured in police action during the march to Parliament on Monday (July 20), called by the Cockroach Janta Party (CJP) sustained pellet gun injuries. The victim identified as Shaikh Irshad Mansoori (25), a resident of Gurugram, underwent surgery at a government hospital to remove the pellets lodged under his eye. Most pellets were lodged in the upper part of his body.

Another video of a woman being stunned by a "shock baton," also emerged on Tuesday (July 21). The use of pellet guns and shock batons against civilian protesters, appears to be the first in Delhi.

What are pellet guns?

Pellet guns, whose technical name is 'pump action guns', are a form of non-lethal crowd-control method used by police and military worldwide. The other popular measures are tear gas, water cannon, pepper spray and taser guns among others.

Pellet guns are also popular in hunting and pest control. Intended to immobilise an individual, they are effective over short ranges of up to 500 yards but when fired from close quarters can be lethal, particularly when sensitive parts like the eyes are hit. The pellets can penetrate soft tissues.

What happened during the July 20 protest in Delhi?

There are standing instructions to open minimal fire at protesters and if necessary "below their waist." A cartridge of a pellet gun contains a few hundred pellets made of lead, which disperses a few hundred metres when fired. Pellets can be of various shapes either like ball bearings or irregular. They are manufactured at the Ordnance Factory, Ishapore.



While Delhi Police has denied its use against protesters, pellet guns are part of the RAF's gear. ANI

While the Delhi Police have denied its use against protesters, pellets guns and shock batons are part of the gear of the Rapid Action Force (RAF) – the anti-riot unit under the Central Reserve Police Force (CRPF) – which was deployed alongside police personnel on July 20 and continues to be deployed near the Jantar Mantar protest site.

On Monday, the victim Shaikh Irshad Mansoori was hit near Palika Bazaar by a RAF official. His friend said that there are around 30-40 injuries on his face and across his body. Four of them, including one in the neck, had penetrated deeply, which is why he underwent surgery.

Originally from Jabalpur in Madhya Pradesh, Mr. Mansoori works at a private company in Gurugram and was attending the protests in solidarity. He also had some visa-related work in New Delhi, which he was supposed to finish after the protest.

What is the first known usage of pellet guns?

Pellet guns were first deployed in Jammu

and Kashmir after Kashmir Valley's summer unrest of 2010 that claimed the lives of 112 protesters in police firing. The then Congress-led government at the Centre deployed pellet guns as a "non-lethal" measure to avoid civilian fatalities.

After violent protests broke out in the Kashmir Valley post July 8, 2016 – when the Hizbul Mujahideen (HM) commander Burhan Wani was killed in an encounter with security forces – several people were blinded due to injuries inflicted by pellet guns, used by the CRPF as a crowd control measure.

Why were its specifications changed?

In 2016, the BJP-PDP government faced severe criticism for using pellet guns, with several political parties asking for its complete withdrawal. Due to the use of pellet guns, 13 people were killed in 2016 and more than 250 were injured, while some lost their vision.

In 2017, a year later, the first batch of less lethal "plastic bullets" were dispatched to the Kashmir Valley. The

plastic bullets were tested at the Terminal Ballistics Research Laboratory (TBRL) in Chandigarh and could be fired from the weapons that the security forces already possessed. Instead of metal, this round of ammunition had a plastic head.

In 2016, 1,022 CRPF personnel were also injured in stone-pelting by protesters. Of these, 956 were injured in the Valley between July 8 and 16 during the unrest.

What is the last definitively known use of pellet guns in the country?

They were last known to have been used during the 2024 farmers protest at Khanauri and Shambhu barriers along the Punjab-Haryana border. Though the police denied using pellet guns, farmer leaders had alleged that several persons were injured.

In 2023, they were used against protesters in the ethnic-strife torn Manipur, leading to a massive public outrage.

What about shock batons?

Shock batons are also a part of the RAF personnel gear.

In 2021, the Ministry of Home Affairs (MHA), through the Border Security Force (BSF), prescribed Qualitative Requirements (QRs) for handheld 'Shock Batons' intended for self-defence and operational use by security personnel. It said that the equipment should be light weight, with a maximum weight of 1 kg and a length of 450 mm to 700 mm, ensuring ease of carriage and handling. It must be constructed from high-impact resistant material, capable of functioning reliably in temperatures ranging from -20°C to +50°C and offering a minimum level of water resistance.

The shock baton should have a service life of at least five years and be powered by a rechargeable battery. Additional requirements include the capability to deliver at least 1,000 quarter-second shock bursts on full charge, thereby ensuring sustained operational effectiveness in field conditions.

THE GIST

At least one of the 80 protesters injured in police action during the march to Parliament on Monday, called by the Cockroach Janta Party (CJP) sustained pellet gun injuries.

Pellet guns were first deployed in Jammu and Kashmir after Kashmir Valley's summer unrest of 2010 that claimed the lives of 112 protesters in police firing.

Are Pellet Guns Being Used at Protests?

● Non-Lethal Crowd Control Weapons

○ A category of weapons designed to incapacitate rather than kill, including tear gas, water cannons, pepper spray, taser guns, and pellet/pump-action guns though “non-lethal” does not mean risk-free, especially at close range

● Rapid Action Force (RAF)

○ A specialised wing of the Central Reserve Police Force (CRPF), raised in 1992 for riot control and communal violence situations, equipped with less-lethal weapons including pellet guns and shock batons

Maintain Status Quo on Water Release, Says Cauvery Body

Maintain *status quo* on water release, says Cauvery body

CWMA directs Tamil Nadu and Karnataka to use present storage only for drinking purposes; it takes note of poor rainfall in basin, says Karnataka released only 0.54 tmcft of water in July so far

B.S. Satish Kumar
BENGALURU

Taking a serious note of poor rainfall in the Cauvery basin, the Cauvery Water Management Authority (CWMA), which met in Delhi on Wednesday, has directed both Karnataka and Tamil Nadu to strictly confine use of their storage to drinking water purposes.

The CWMA, which is the apex body that monitors implementation of Cauvery water release to lower riparian States as per the Supreme Court's verdict, directed both the States to maintain *status quo* with respect to water release and utilisation.

According to sources,



Trying times: The water level in the KRS dam across the Cauvery in Karnataka is fast depleting. SPECIAL ARRANGEMENT

the CWMA referred to the IMD forecast that indicates further tough time in terms of rainfall in the coming days, and observed that it is going to be a distress year for the Cauvery basin.

The present storage in

reservoirs of the Cauvery basin in both Karnataka and Tamil Nadu could only take care of their drinking water requirements, the authority reportedly observed.

Warning both the States

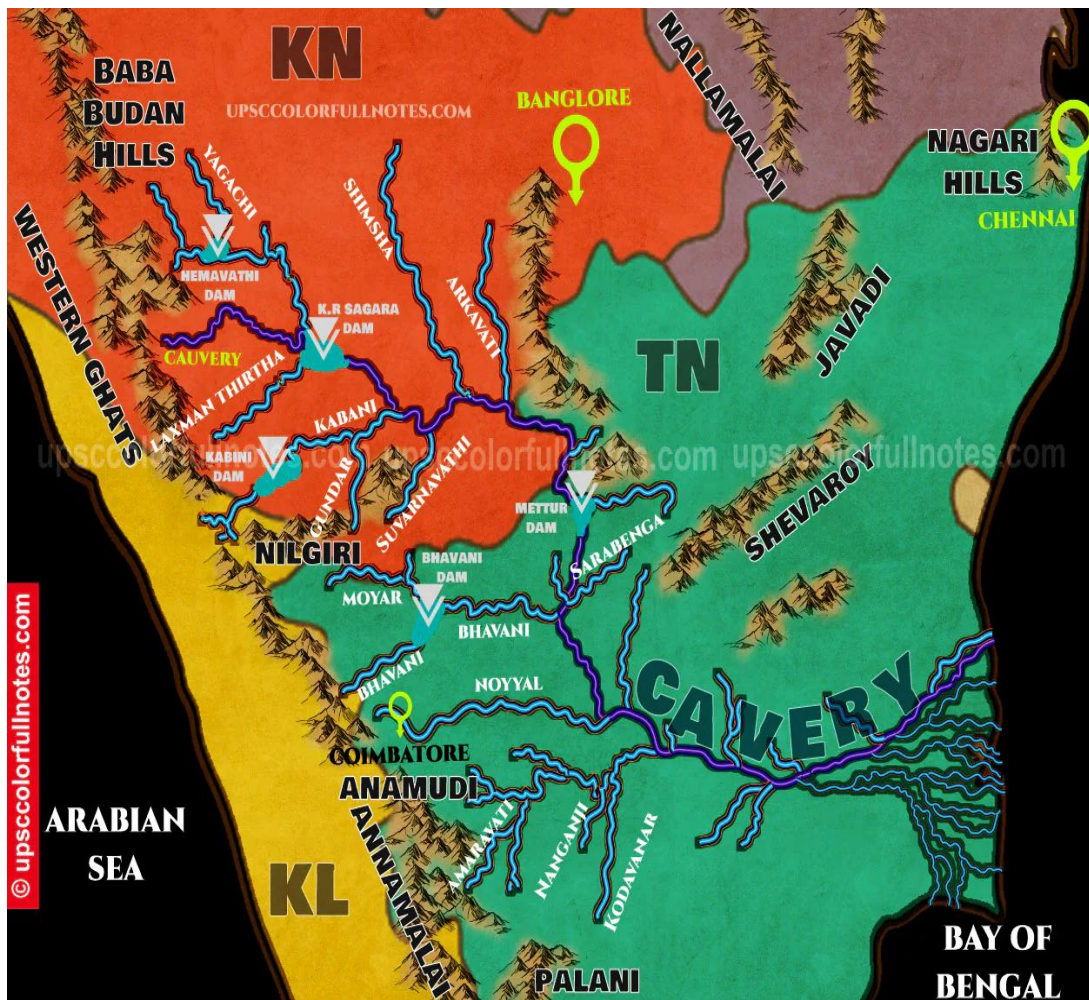
against using the water storage for other purposes including irrigation, the authority expressed confidence that it would be possible to prevent any drinking water shortage if the storage in reservoirs is judiciously used, sources said.

At the meeting, it was also noted that Karnataka had released only 2.5 tmcft of water in June and 0.54 tmcft during July so far to Tamil Nadu, which is lower than that prescribed.

Review on July 28

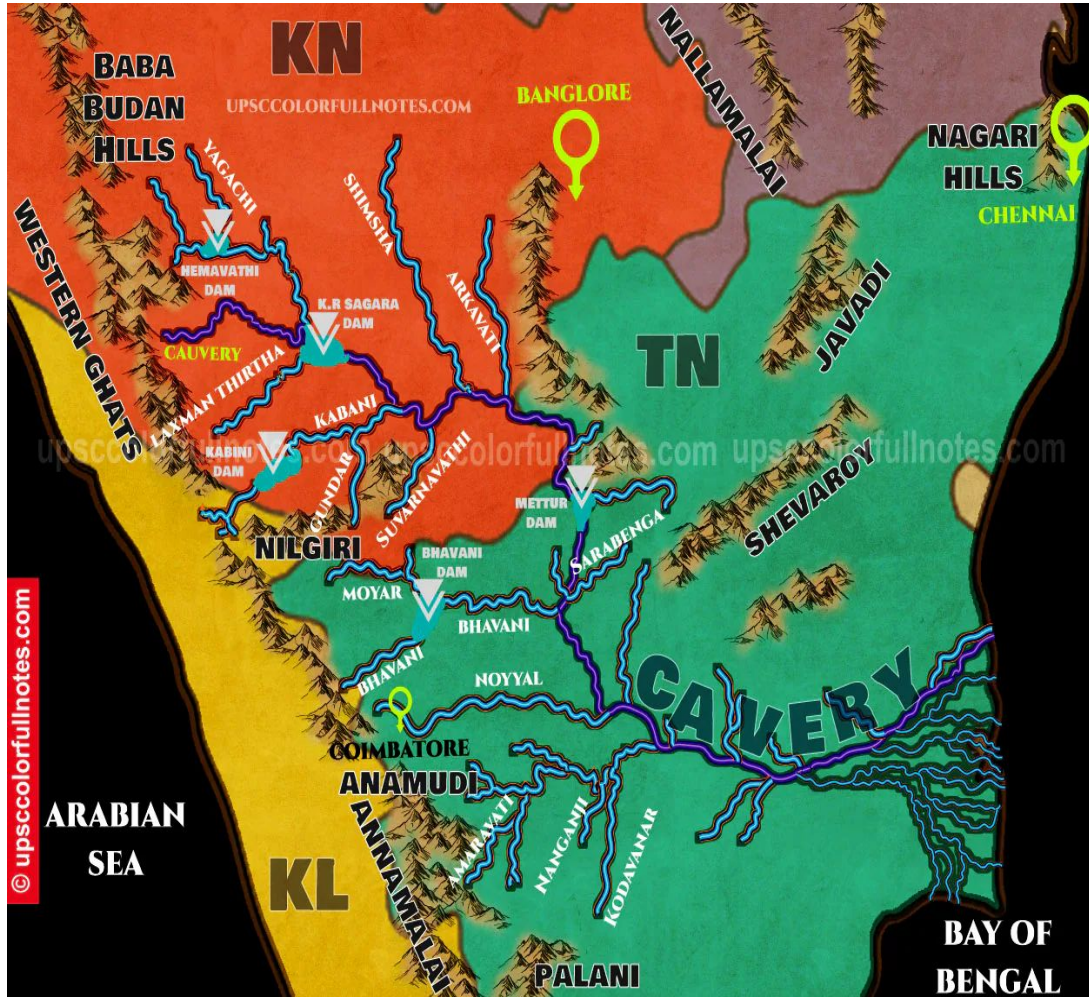
The situation will be reviewed on July 28 by the Cauvery Water Regulation Committee, which assists the CWMA in monitoring the storage, releases and other issues.

Maintain Status Quo on Water Release, Says Cauvery Body



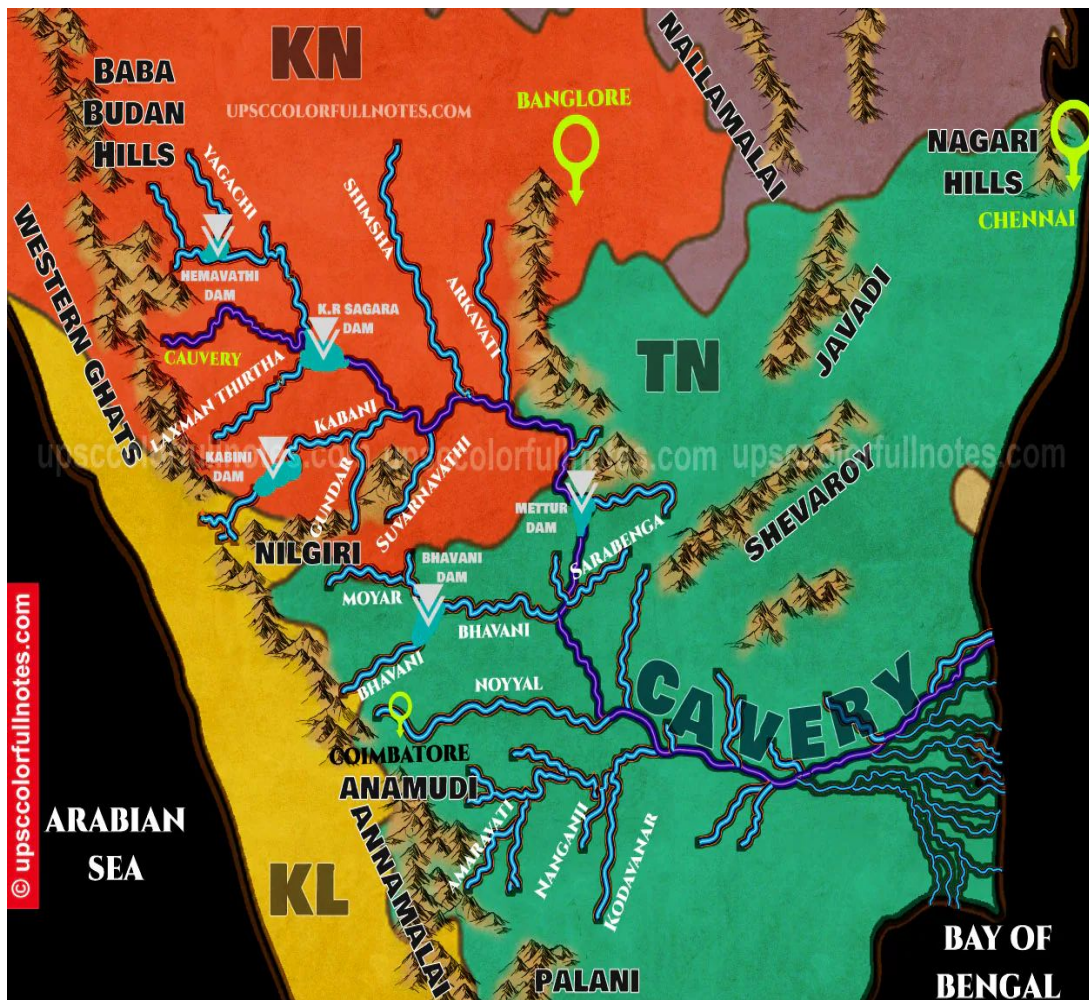
- The Cauvery River (Kaveri) is designated as the ‘Dakshi Bharat ki Ganga’ or ‘the Ganga of the South’.
- The Cauvery basin extends over states of Tamil Nadu, Karnataka, Kerala, and Union Territory of Puducherry draining an area of 81 thousand Sq.km.
- The Cauvery River rises at an elevation of 1,341 m at Talakaveri on the Brahmagiri range near Cherangala village of Kodagu (Coorg) district of Karnataka.
- The Hemavathi Dam is a large masonry dam located in Karnataka, India, on the Hemavathi River, a tributary of the Kaveri River. It is a critical facility for irrigation, drinking water supply, and hydroelectric power generation, serving principally the Hassan.
- Laxman Tirtha River : It is a tributary of the river Kaveri in Karnataka. It originates in the Brahmagiri hills, Kodagu or Coorg District, Karnataka, and flows eastward. It joins the Kaveri River in the Krishna Raja Sagara Lake
- Krishna Raja Sagar Dam : The reservoir is also the main source of drinking water for all of Mysore city and almost the whole of Bangalore.

Maintain Status Quo on Water Release, Says Cauvery Body



- The Kabini River is one of the major tributaries of the river Cauvery in southern India, It originates in Wayanad district of Kerala state by the confluence of the Panamaram River and the Mananthavady River.
- Suvarnavathi is another right bank tributary
- Shimsha and Arkavathi river are another left bank tributary
- At the confluence of Kaveri and Arkavathi river is the proposed Mokedatu project
- Hogenakkal Waterfalls – The Niagara of India is located on the Karnataka and Tamil Nadu Border
- Mettur Dam : It is located at Mettur, Salem District, in Tamil Nadu.

Maintain Status Quo on Water Release, Says Cauvery Body



- **Bhavani River :** It is a major tributary of the Cauvery River. It flows through the states of Kerala and Tamil Nadu. It is the second largest river in Tamil Nadu. The Bhavani River starts its journey in the Nilgiri hills, which are part of the Western Ghats mountain range. From there, it flows into Kerala, passing through the beautiful Silent Valley National Park. After flowing through Kerala, the river turns and flows back into Tamil Nadu.
- Noyal and Amaravati are two tributaries in the right bank
- It then passes through Tiruchirappalli
- The Grand Anicut, also known locally as Kallanai, is one of the oldest and most remarkable irrigation structures in the world. Located on the Cauvery (Kaveri) River in the Thanjavur district of Tamil Nadu, India, it is a stone masonry dam built during the reign of the Chola dynasty, traditionally attributed to the great Chola king Karikala Chola in the 2nd century CE.

Maintain Status Quo on Water Release, Says Cauvery Body

Basin

Tamil Nadu (56%) , Karnataka (41%), Kerala (3%)

Which one of the following statements is **not** correct? (2006)

- (a) Mahanadi River rises in Chhattisgarh
- (b) Godavari River rises in Maharashtra
- (c) Cauvery River rises in Andhra Pradesh
- (d) Tapti River rises in Madhya Pradesh

Maintain Status Quo on Water Release, Says Cauvery Body

Consider the following pairs:

<i>Famous Place</i>	<i>River</i>
1. Pandharpur	Chandrabhanga
2. Tiruchirappalli	Cauvery
3. Hampi	Malaprabha

Which of the pairs given above are correctly matched?

This question was previously asked in
Official UPSC Civil Services Exam
2019 Prelims Part A

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1. 1 and 2 only

2. 2 and 3 only

3. 1 and 3 only

4. 1, 2 and 3

Supreme Court-Appointed Aravali Panel's Consultation Process Excludes Rural People

Supreme Court-appointed Aravali panel's consultation process excludes rural people, say environmentalists

Press Trust of India
NEW DELHI

Environmentalists and activists have raised concerns about the Supreme Court-appointed committee inviting written feedback on issues related to the Aravali hills and ranges through e-mail or Google forms specifically designed for the purpose.

They have argued that the online outreach mechanism excludes rural communities, the most directly impacted by decisions on the Aravalis and which may not be literate and technology-savvy.

Formed on May 25, the high-powered committee has been tasked with conducting an independent review of the Centre's report on the definition and



Fragile nature: There has been a demand that the committee members visit the villages in the Aravali belt. FILE PHOTO

delineation of the Aravali hill range. The committee has to submit its report by August 31.

'Not for rural people'

"The online submission process announced by the high-powered committee is not designed for rural people. A villager in the Aravali belt cannot navi-

gate a complex e-mail address or Google Form, let alone ensure it reaches the right inbox without a typographical error," said Kusum Rawat, member of the pan-India Adivasi Samanvay Manch Bharat.

Ms. Rawat added that the committee should paste public notices in local languages at panchayat

bhavans, community halls, government schools, and anganwadis in Aravali districts across five States.

There has also been a concern about the 21-day window for sending feedback as environmentalists and activists claim that the time period is not sufficient to communicate concerns in a meaningful way.

'Short deadline'

"Twenty-one days is an entirely inadequate window to hear from the diverse communities living deep within the Aravali belt. Getting these communities to participate in any meaningful way will be a manual, time-intensive process – there are no shortcuts and decent coverage of voices within this

timeline is simply not possible," Lokesh Bhiwani, founder of Stand With Nature, said.

There has also been a demand that the committee's members visit the villages in the Aravali belt.

'Flouting of rules'

"Most of the licensed mining leases running today across the States of Haryana, Rajasthan, and Gujarat are blatantly flouting rules and norms, causing deep damage to the health and livelihoods of local communities. No written submission can capture the everyday suffering being endured by the people dependent on the good health of the Aravalis for their existence," said Diwan Singh, convener, Ridge Bachao Andolan.

Supreme Court-Appointed Aravali Panel's Consultation Process Excludes Rural People

Aravalli Range

The Aravalli range is among the world's oldest living mountain systems and the geological spine of northwest India.

Mountain Type: It is an ancient fold mountain range now eroded into residual hills with average elevations of 300-900 metres.

Geographic Extent: It stretches about 692 km from Champaner in Gujarat, through Rajasthan and Haryana, to Raisina Hill in Delhi.

Highest Peak: Guru Shikhar, at 1,722 metres on Mount Abu in Rajasthan, is the highest point.

Thank You